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Deep learning based detection and localization of road accidents from traffic surveillance videos

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Abstract

Real-world traffic surveillance videos need continuous supervision to monitor and take appropriate actions in case of fatal accidents. However, continuously monitoring them with human supervision is error prone and tedious. Therefore, a deep learning approach for automatic detection and localization of road accidents has been proposed by formulating the problem as anomaly detection. The method follows one-class classification approach and applies spatio-temporal autoencoder and sequence-to-sequence long short-term memory autoencoder for modeling spatial and temporal representations in the video. The model is executed on a real-world video traffic surveillance datasets and significant results have been achieved both qualitatively and quantitatively.
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Keywords: Accident detection; Deep learning; One-class classification; Video surveillance

1. Introduction

The proliferation of Artificial Intelligence and the ubiquitous use of Internet of Things devices have resulted in smart city projects for improving the quality of life [1,2]. Sustainable transportation is one of the crucial dimensions in smart cities which aims to develop (1) intelligent transportation system for public sector relying on real-time data, (2) traffic management platforms to avoid the congestions, and (3) safety and proactive traffic control applications. The increase in number of vehicles on the road causes much burden on road traffic management personnel and authorities to handle challenges like road accidents, thus, requiring quick surveillance to mitigate the chances of the loss of lives and properties. Therefore, for detecting and avoiding the mishaps like accidents, cities are deployed with CCTV surveillance cameras for traffic monitoring purposes. Manually checking the surveillance videos in real time for monitoring the traffic and identifying the occurrence of any unwanted incidents by human personnel is not a feasible solution. Moreover, the videos are continuously streamed from CCTV cameras and some unusual activities may get missed by error-prone manual surveillance. This necessitates devising an automated model for detecting unusual incidents like accidents for sustainable transportation and effective traffic management.

As a panacea, computer-vision based techniques have been popularly used for intelligent traffic surveillance and management. The general pipeline of computer vision based techniques follows (1) Segmentation of vehicles from the background and other road entities from still as well as moving video scene (2) Classification of all the categories of vehicles such as bus, van, car, etc. (3) Extraction of spatio-temporal features for different traffic related operations like vehicle counting, vehicle tracking, trajectory tracking, detection of anomalous events like accidents. Moreover, such computer vision based systems should operate under varying lighting and traffic conditions. Motivated by the same, a deep learning approach for detection and localization of road accidents has been proposed. Specifically, we consider occurrence of road accident as a special case of unusual activity i.e., anomaly. Detecting and localizing the road accidents would help to reduce the response time of rescue teams and enable travel safety.

The contributions of the paper are stated as follows.

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- This paper addresses an issue of detecting and localizing the road accidents as a special case of anomaly detection by using deep learning approach based on spatio-temporal autoencoder and sequence-to-sequence long short-term memory (LSTM) autoencoder.
- To handle the issue of scarcity of labeled anomalous data, the model needs to be trained on normal events only and then it detects anomalies as out-of-distribution data by following one-class classification paradigm.

The research proposed in this paper addresses the task of accident detection by following unusual activity detection approach based on deep learning and one-class classification paradigm. To the best of our cognizance, we are the first to apply the variant of LSTM, namely sequence-to-sequence LSTM for the task of anomaly detection in a sliding window manner justifying the novelty. The proposed model applies convolutional autoencoder and LSTM with sequence-to-sequence learning ability in a cascaded manner. The crux of the proposed approach is that it learns in an unsupervised manner alleviating the need to generate labeled anomalous road accident data.

The contents of the paper are delineated as follows. Section 2 discusses approaches addressing the task of road accident detection. How the task of accident detection has formulated as an anomaly detection problem and the detailed methodology have been described in Section 3. Section 4 sheds light upon experimentation details, datasets and performance metrics used for evaluating the proposed model along with quantitative and qualitative results and related comparative analysis. Section 5 portrays the concluding remarks and pointers to extend the current research work.

2. Related work

There has been a lot more research conducted in the domain of visual surveillance applied to road accident detection. A detailed review on anomaly detection in road traffic accidents is extensively covered in [3] giving pointers to address issues in Intelligent Transportation Systems. By and large, the task of accident detection for traffic video surveillance has been formulated in terms of techniques based on traffic flow patterns, vehicle tracking and interaction of vehicles, spatio-temporal correlation, object detection and unusual activity detection as shown in Fig. 1.

The first category of approach is based on traffic flow patterns which generally model the normal traffic patterns such as valid left turns, right turns or going straight. Deviation from normal patterns is considered as anomalous traffic event [4,5]. The second category of approach focuses on detecting and extracting the features of moving vehicles such as distance between vehicles, acceleration, direction, tracking [6]. Hui et al. [6] used Gaussian Mixture Model for vehicle detection and mean shift algorithm for vehicle tracking. The third category of approach follows sociological concepts and model interaction between the vehicles for detection of accidents. Sultani and Choi [7] proposed an intelligent model in which

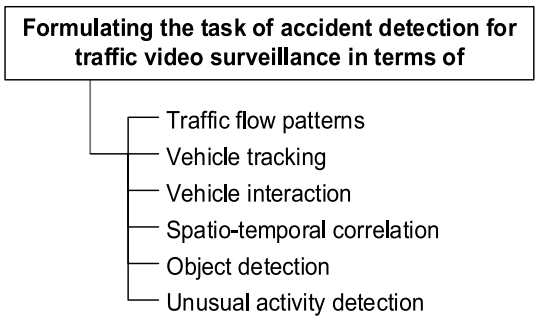


Fig. 1. Techniques for formulating the task of accident detection for traffic video surveillance.

interaction among the cars is considered as a guiding point for anomaly detection. The traffic is modeled using Latent Dirichlet Allocation (LDA). For localization, spatial gradients of behavior are computed to construct Finite Time Lyapunov Field and Watershed algorithm is used to localize the anomalies. Due to large availability of data and computing processors at cheaper costs, deep learning technology has been resurged and proved its remarkable usability in many domains like object detection [8,9], image classification [10], anomaly detection [11], topic modeling [12], representation learning [13], semantic segmentation [14], visual tracking [15], etc. and so deep learning techniques have also been used for accident detection.

Ren et al. [16] studied the effect of spatio-temporal correlation in the traffic accident data and developed recurrent neural network based model for accident risk prediction which can act as a proactive system to warn regarding the traffic accidents. For improved accuracy, more features like characteristics of road, traffic flow can also be incorporated. Bortnikov et al. [17] implemented 3D Convolutional Neural Network (CNN) based accident detection model addressing different weather and lighting conditions. They used video game for generating video traffic and training the model. During testing, they applied two loss functions with and without optical flow patterns to curb the effect of illumination conditions and background scenes. The model is evaluated on real-world traffic videos.

Tian et al. [18] proposed a deep learning model for car accidents based on object detection algorithm — You Only Look Once (YOLO) and utilized other information obtained from vehicle infrastructure system such as accident location, accident type and lighting conditions. This model leverages multi-scale feature fusion technique and loss function having dynamic weights for accurate and real-time detection. The approach proposed by Yao et al. [19] follows techniques based on both object tracking and unusual activity. They applied unsupervised learning approach for dashboard-mounted camera videos. In this, a model is trained with normal events of objects and ego motions. While testing, the patterns which deviate from the normal model, are regarded as anomalies. Specifically, their model predicts location of objects in the

ongoing frame based on previous frames and then applies anomaly detection algorithm.

Apart from this, there are also the deep learning approaches addressing visual surveillance for road traffic and road surface monitoring [20,21]. Ohgushi et al. [20] proposed to address the task of detecting obstacles on road using autoencoder with semantic segmentation. Specifically, autoencoder consists of semantic image generator as encoding part and photographic image generator as decoder. Their approach works in an unsupervised manner such that model is trained on normal road scenes only. Varona et al. [21] proposed to use deep learning models based on convolutional neural networks, LSTM networks, and reservoir computing models for pothole detection and road surface classification.

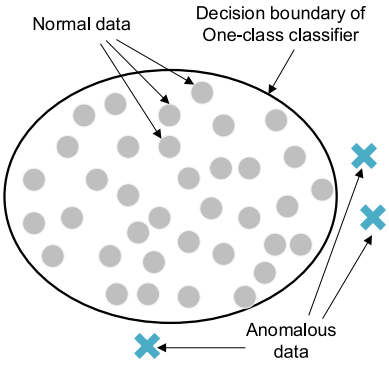


Fig. 2. One-class classification.

3. Proposed methodology

3.1. Problem formulation

We formulate the task of traffic road accident detection as an anomaly detection problem. Anomalous patterns deviate from normal patterns and anomalies occur for short duration of time. However, due to dearth of labeled data for anomalous activities, a model is trained on normal events only and then tested on anomalous data i.e., a model learns normal data patterns and patterns which do not conform to normal one are regarded as anomalous. This paradigm is called as one-class classification and it is depicted in Fig. 2. Learning the normal patterns results into formation of some hypersphere and the points which lie outside such hypersphere are anomalous. Generally, in one-class classification problems, normal data is assumed to be a negative class and anomalous data to be a positive one [22].

3.2. Proposed methodology for anomaly detection and localization

Autoencoders are very useful for efficient representation of videos. Therefore, our previous work [13] focused on rigorous analysis of autoencoders and accordingly, convolutional autoencoder and LSTM autoencoder with sequence-to-sequence learning capability have been used for the task of spatio-temporal feature learning in this research. Fig. 3 depicts the working of the proposed model for anomaly detection in traffic surveillance videos. The details of convolutional autoencoder and LSTM autoencoder with sequence-to-sequence learning capability have been depicted in figures 4 and 5 respectively. Figs. 4(a) and 5(a) give the layerwise details and activation functions used in the encoder part of convolutional autoencoder and LSTM autoencoder with sequence-to-sequence learning capability respectively. Similarly, Figs. 4(b) and 5(b) give the details of decoder part.

A sequence of video frames $\{f_0, f_1, \ldots, f_n\}$ forms a video clip denoted by $f_{0\ldots n}$. A convolutional encoder, $\mathcal{CAE}$ (.) is used for getting the encoded representation of each frame in the video clip for spatial learning. $\mathcal{CAE}$ (.) uses $\mathcal{L}_2$ distance to minimize the error between actual frame and the reconstructed frame. The latent representation of each video frame outputted by $\mathcal{CAE}$ (.) is stacked together in a window size of m by applying the sliding window technique. This stack of latent representations is fed to LSTM autoencoder with sequence-to-sequence learning capability i.e., $\mathcal{SLAE}$ (.) for learning the temporal patterns. To follow one-class classification, we fit the normal frames (training data) to a Gaussian distribution. To achieve this, radial basis function layer is applied before the bottleneck layer in LSTM encoder with sequence-to-sequence learning $\mathcal{SLAE}^{enc}$ (.). Due to this, a hypersphere is formed by fitting a normal data to a Gaussian distribution. During testing, out-of-distribution (anomalous) data falls outside such hypersphere, helping to detect the anomalies. Once training data is fitted to a Gaussian distribution with RBF kernel, η dimensional vector is generated for a stack of m frames obtained using sliding window technique. Once training gets completed, all sequences ($f'_{0\ldots n}$) are collected having dimensions as $[\mathcal{K}, \eta]$ where $\mathcal{K}$ denotes number of training sequences and η represents the dimension of the latent vector. Then, for each dimension η of the latent vector, mean (μ_i) and standard deviation (σ_i) are calculated as follows.

$$\mu_i = \frac{1}{\mathcal{K}} \sum_{j=1}^{\mathcal{K}} f'_{0\ldots n}[j][i] \tag{1}$$

$$\sigma_i = \sqrt{\frac{\sum_{j=1}^{\mathcal{K}} (f'_{0\ldots n}[j][i] - \mu_i)^2}{\mathcal{K}}} \tag{2}$$

After calculating μ_i and σ_i, algorithm 1 is applied for detecting whether a test sequence, $f^{test}_{0\ldots n}$ is anomalous or not. For getting the anomaly score, hyperparameters α and β have been used. Algorithm 1 shows the steps applied for detection of anomalies. For anomaly localization, a moving window of $h \times w$ is taken where h denotes height and w denotes width, and then slid through the entire reconstructed frame obtained from convolutional decoder. If a particular window is more than $\beta \times \eta$, then that window is annotated as anomalous and bounding box is applied.

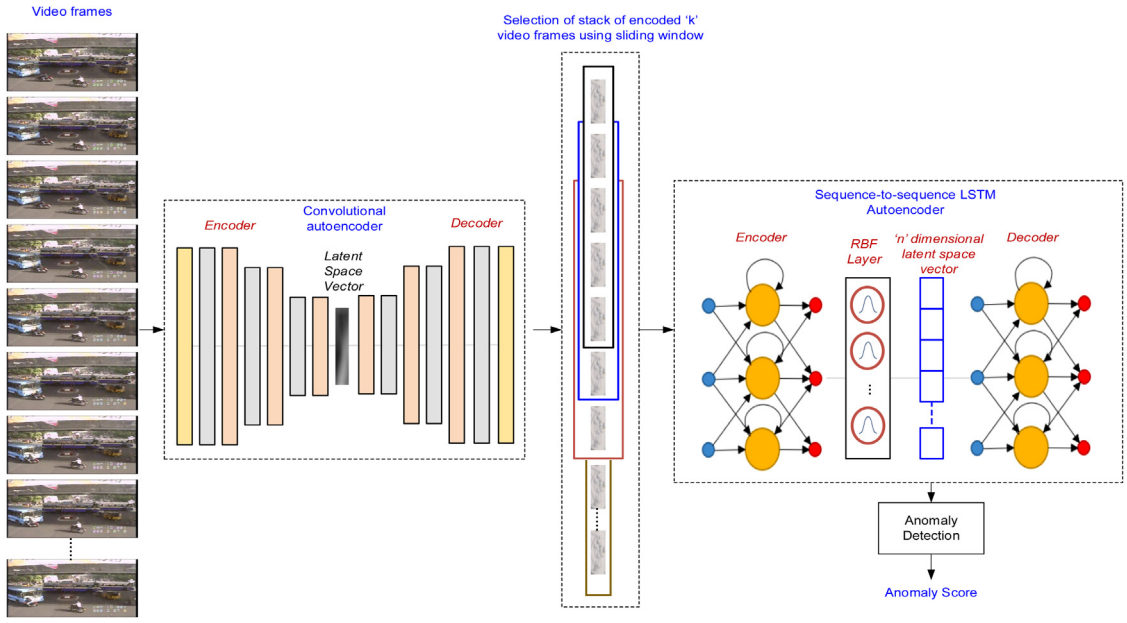


Fig. 3. Proposed model for anomaly detection in traffic surveillance videos.

Algorithm 1: Anomaly Detection from videos
Input: *Sequence of encoded video frames* - $f_{0\ldots n}^{test}$, *hyperparameters* - α *and* β, *dimension of latent space* $-\eta$
Output: Anomaly score

ANOMALY DETECTION $(f_{0\ldots n}^{test}, \alpha, \beta, \eta)$
1. $score \leftarrow 0$
2. **for** i$\leftarrow 1, \eta$
3. **if** $(f_{0\ldots n}^{test}[i] - \mu_i) \geq \alpha \cdot \sigma_i$
4. $score \leftarrow score + 1$
5. **if** $score \geq \beta \cdot \eta$
6. **return** "anomalous"
7. **else**
8. **return** "not anomalous"

4. Experimentation

Hardware and software. For conducting the experiments, machine with NVIDIA GeForce GTX 940MX GPU with 4 GB RAM and i7-7500U CPU 2.9 GHz and having NVIDIA GeForce GTX 940MX GPU and 4 GB RAM been used. TensorFlow 2.0.0 framework and Python 3.7.5 are used for implementation.

4.1. Experimentation on IITH road accident dataset

4.1.1. Details of IITH road accident dataset

The effectiveness of the proposed model is tested on real-world traffic surveillance video, namely, IITH road accident dataset [23]. This dataset has been collected from the network of CCTV cameras installed at different squares capturing the scenes of road accidents in the city of Hyderabad, India. Fig. 6 shows sample videos frames. Every scenario in the dataset commences with normal traffic events and ends with the event of road traffic accident. The details of the dataset are given in Table 1.

Table 1
Details of IITH road accident dataset.

Frames	Count of frames
Normal frames	1,27,138
Anomalous frames	863
Normal frames used for training	94,720
Frames used for testing	33,281 (32,418 normal frames + 863 accident frames

4.1.2. Evaluation metrics

Equal Error Rate (EER). EER can be defined as follows.

$$EER = \frac{FP + FN}{n} \tag{3}$$

where, FP denotes false positive, FN denotes false negative and n denotes total count of video frames under consideration for anomaly detection. A lesser value of the EER implies that model is better in detecting the anomalies.

Area Under the Receiver Operating Characteristics Curve (AUC). AUC value is obtained by first plotting the ROC curve and calculating the area covered by that curve. ROC curve shows the classification ability of the model by plotting the FPR and TPR at all classification thresholds. A higher value of AUC indicates that model is better.

Mean Average Precision (mAP). In mAP, each threshold represents percentage of overlap between predicted bounding box and ground truth bounding box. This threshold is called as Intersection over Union (IoU). It is calculated as follows.

$$IoU = \frac{Area\ of\ overlap}{Area\ of\ Union} \tag{4}$$

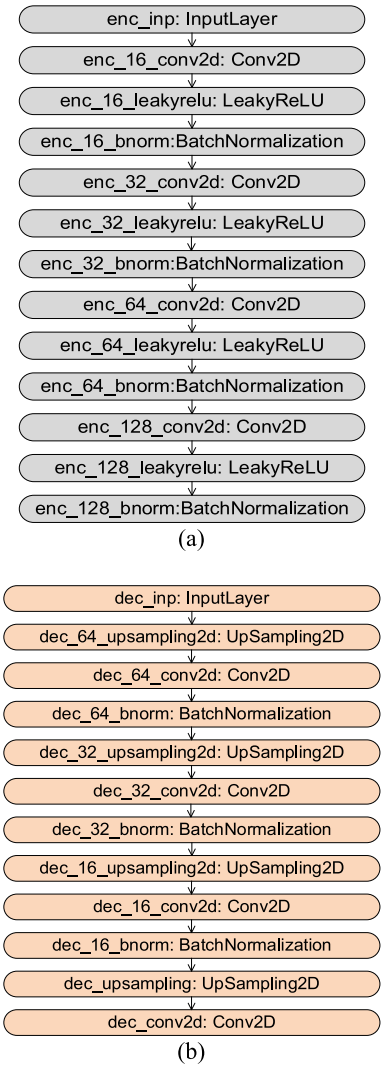


Fig. 4. Details of convolutional autoencoder (a) Encoder (b) Decoder.

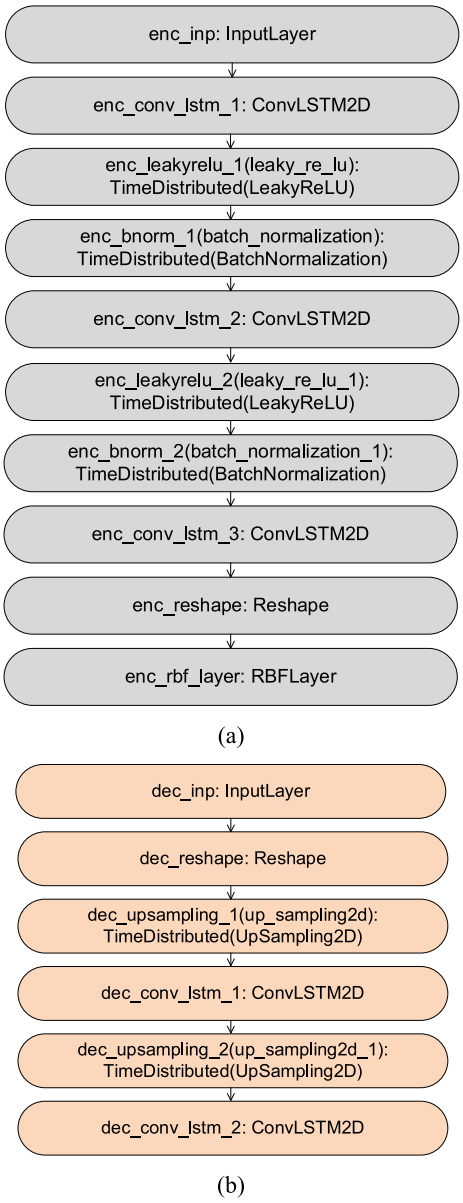


Fig. 5. Details of LSTM autoencoder with sequence-to-sequence learning capability (a) Encoder (b) Decoder.

The higher the value of mAP implies model is better in localizing the regions of interest.

4.1.3. Results discussion and comparative analysis

For checking the performance of the model at frame-level, i.e., checking whether a model is correctly identifying whether a frame has an anomaly or not, EER and AUC have been used. Similarly, for evaluation at pixel-level, i.e., checking whether a model is correctly localizing the anomaly in the frame, we used mAP as the performance metric. For frame-level detection, the ROC curve has been obtained as shown in Fig. 7 and corresponding AUC is 79%. An EER of 20.50% has been achieved for anomaly detection at frame-level. As IITH road accident dataset does not provide ground-truth for checking the effectiveness of the model at pixel-level (localization), we manually annotated dataset with bounding box coordinates in each of our extracted video frame using an annotation tool labellmg and then performance is checked using mAP. The mAP values corresponding to an IoU threshold range ranging from 0.1 to 1.0 have been shown in Fig. 8. At 40% threshold, the proposed model achieved 60% of mAP for pixel-level evaluation as shown in Fig. 8. Figs. 9 and 10 show the qualitative performance of the model on IITH road accident dataset in which accident scenes are localized by the model.

We compared the performance with results of the paper by Singh and Mohan [24] as shown in Table 2. This is a single paper available in the literature in which an evaluation is performed at frame-level on IITH Road accident dataset. Based on the comparison made in Table 2 the proposed model achieved 2% of improved performance in terms of AUC over state-of-the-art method [22].



Fig. 6. Sample video frames from IITH road accident dataset [23].

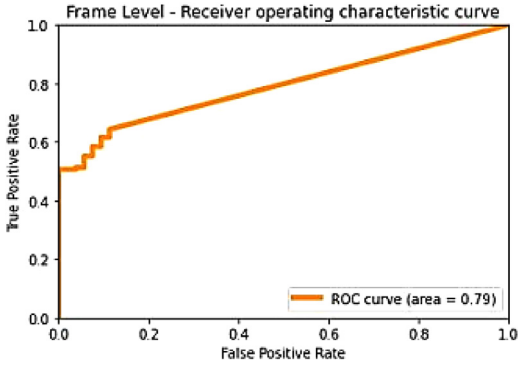


Fig. 7. ROC at frame-level evaluation (AUC = 79%).

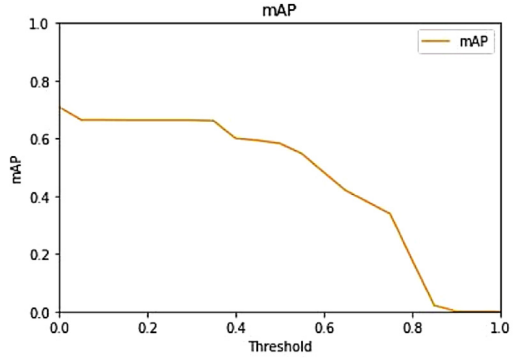


Fig. 8. mAP scores with varying threshold.



Fig. 9. Localization showing a bike colliding with a bus.



Fig. 10. Localization showing an auto-rickshaw striking with a car.

Table 2
Comparative analysis for IITH Road accident dataset.

Method	Frame-level AUC	EER	mAP
Stacked autoencoder [24]	77%	22.50%	–
Proposed model	79%	20.50%	60%

4.2. Experimentation on Iowa DOT anomaly dataset

4.2.1. Details of Iowa DOT anomaly dataset from 2021 AI City Challenge

2021 AI City Challenge is divided into 5 tasks *viz.* Multi-Class Multi-Movement Vehicle Counting Using IoT Devices, City-Scale Multi-Camera Vehicle Re-Identification, City-Scale

Multi-Camera Vehicle Tracking, Traffic Anomaly Detection, and Natural Language-Based Vehicle Retrieval [25]. Out of the five challenges, we executed the proposed model on 2021 AI City Challenge dataset belonging to track 4 of traffic anomaly detection [26]. This dataset is collected by DOT of Iowa captured at multiple freeways. The anomalies include crashes of single and multiple vehicles as well as occurrence of stalled vehicles. There are 100 and 150 videos in training and test set respectively in traffic anomaly detection dataset (Iowa DOT anomaly dataset). The video clips have been recorded at 30 frames per second with a resolution of 800 × 410. The duration of the video clips in this dataset is of about 15 min.

As mentioned in Section 3, the proposed model builds a normalcy model based on normal data only and detects road accidents as anomalies (out of distribution samples) using one-class learning. Moreover, the proposed model supports to detect single and multiple crashes of vehicles as part of accident, however, it is not accountable for detecting the stalled vehicles. Therefore, to test the proposed model on traffic anomaly detection dataset of AI City challenge, we curated the dataset in the following manner. We removed video clips belonging to stalled vehicles (which does not show occurrence of road accident) from both training and test set. In addition, video clips having occurrences of accidents (single and multiple crashes of vehicles) have been removed from the training set. Then, the model is trained on normal video clips of the curated dataset and tested on dataset having both occurrence of normal and accident events. This way model is trained in unsupervised manner (training on normal data and testing on normal as well as anomalous road accident data).

4.2.2. Evaluation metrics

To justify the performance of the model in terms of accident detection, F_1 score has been used as mentioned in AI city challenge [25]. F_1 score is computed as follows.

$$F_1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall} = \frac{TP}{TP + \frac{1}{2}(FP + FN)} \quad (5)$$

In Eq. (5), a true-positive (TP) detection indicates a frame is correctly predicted to be anomalous within 10 s of real anomalous event (accident event). A false-positive (FN) indicates that model fails to correctly predict a real anomalous event. A false-positive (FP) indicates that predicted anomalous event is not an actual anomalous event.

4.2.3. Results discussion and comparative analysis

With reference to evaluation metrics by AI City challenge [25], F_1 score is used for assessing the performance of model for accident detection. We achieved 78.58% of F_1 score based on training and testing strategy mentioned in Section 4.2.1. The results have been compared with papers addressing the task 4 of AI City challenge of traffic anomaly detection. However, it can be noted that the methodologies presented in papers [27–31] follow supervised learning strategy in which a model is trained on both normal video clips and accident video clips. On the contrary, the proposed model follows unsupervised learning. Fig. 11 shows the comparison

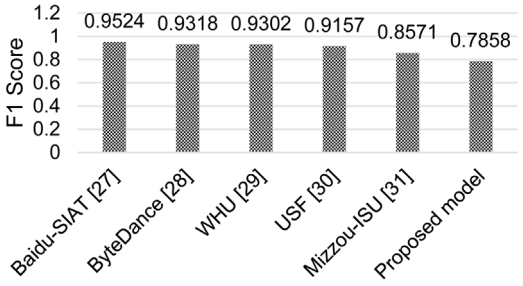


Fig. 11. Comparative analysis for Iowa DOT anomaly dataset based on F_1 score.

of the proposed unsupervised learning approach versus supervised learning approaches from 2021 AI City challenge (track 4) based on F_1 score. Fig. 12 shows qualitative evaluation of the model in terms of localizing the incident of cars crashing into each other from 2 different scenes in Iowa DOT anomaly dataset.

The methodologies adopted for addressing the task 4 of AI City challenge by state-of-the-art [27–31] papers are discussed. Baidu-SIAT [27] follows pipelined framework encompassing pre-processing, dynamic tracking, and post-processing. The pre-processing involves background modeling, vehicle detection, road mask construction and anomalous vehicle tracking. Dynamic tracking models the motion patterns and spatio-temporal tracking of vehicles for locating the start time of anomalies. ByteDance [28] applies background modeling, box-level tracking of spatio-temporal anomalous tubes, and refinements for accurately detecting the time periods of anomalies. WHU [29] uses box-level and pixel-level tracking for identification of the anomalies and dual-modality bilateral trajectory tracing to extract the spatio-temporal features from frames. The methodology of USF [30] involves modeling the background of frames in forward and backward manner, followed by two-stage vehicle detection, and computing the structural similarity for detected regions. Mizzou-ISU [31] follows background modeling, road mask extraction and adaptive thresholding for anomaly detection. To summarize, papers [27–31] involve the usage of YOLO object detector, supervised learning, background modeling and tracking. On the contrary, the proposed approach in this paper follows unsupervised learning strategy and thus, performance on Iowa DOT anomaly dataset varies. However, proposed approach only needs to be trained on precursor video frames prior to anomalous events and works efficiently without any need for labeled anomalous data.

4.3. Experimentation on DoTA dataset

4.3.1. Details of DoTA dataset

Detection of Traffic Anomaly (DoTA) is a traffic anomaly detection dataset collected from YouTube channels having numerous instances of dash camera accident videos [32]. The dataset contains 4677 videos with temporal, spatial, and categorical annotations. The anomalies in the dataset are collision of vehicles on the road due to accidents. For comparison with other state-of-the-art approaches, we followed the strategy

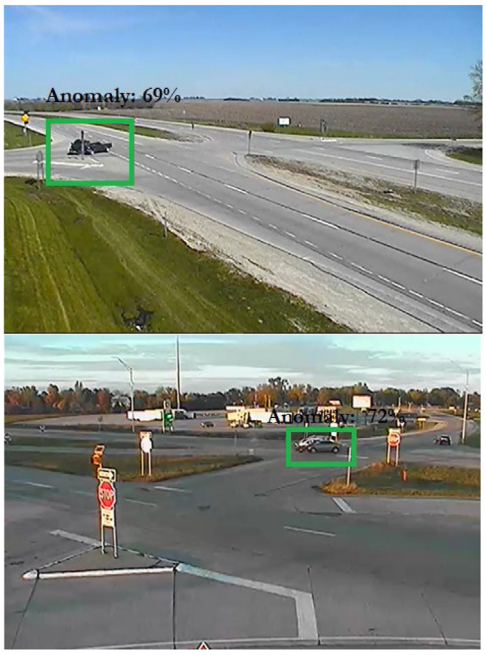


Fig. 12. Localization showing a car crashing into another car.

mentioned in [33]. The dataset has been randomly partitioned into 3275 training and 1402 test videos. As mentioned in Section 3, the proposed model follows one-class learning strategy. So, we extracted precursor normal frames from each video for training and built a normalcy model. Finally, a model is tested on test data for road accident detection.

4.3.2. Evaluation metrics

We evaluated the performance of the model at frame-level using ROC curve and AUC.

4.3.3. Results discussion and comparative analysis

We achieved an AUC of 84.70% for detection of road accidents at frame-level. For comparison, we assessed the performance of the model using AUC against other state-of-the-art unsupervised learning approaches as shown in Fig. 13. A summary of approaches compared is given. In ConvLSTM-MAE [34], a 2D CNN is used for extracting spatial information from frame and multi-layer ConvLSTM encodes the temporal features in a recurrent way and finally 2D decoder is used for reconstructing the input frames. ConvAE [35] method used spatio-temporal autoencoder. It encodes temporally stacked images with 2D convolutional encoders and decodes with deconvolutional layers to reconstruct the input. AnoPred [36] method accepts four continuous RGB images as input and predicts future images with the help of UNet model and uses multi-task loss functionality for improving the accuracy of prediction. TAD [19] method models bounding box trajectories of normal videos with a multi-stream RNN autoencoder. Past trajectories and ego motion are encoded to predict future object boundaries. Ensemble [33] method combines approaches proposed in [19,36] to get advantages of both appearance-only and object-centric methods for detection.

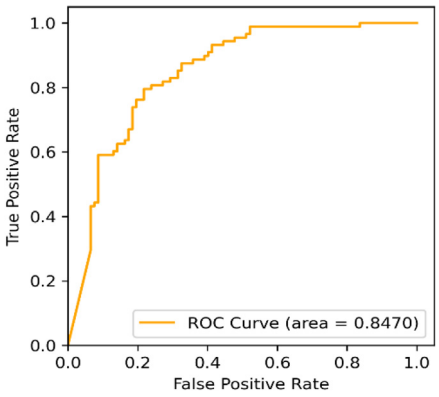


Fig. 13. ROC at frame-level evaluation (AUC = 84.70%).

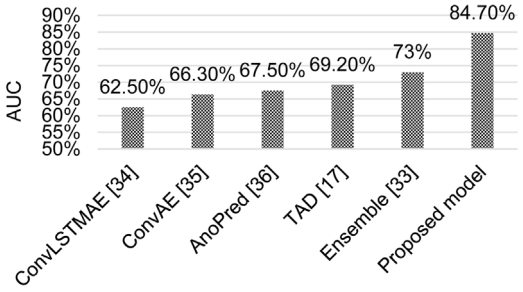


Fig. 14. Comparative analysis for DoTA dataset based on AUC.

Based on the comparison depicted in Fig. 14, the proposed model achieved 11.7% improvement over the state-of-the-art unsupervised learning based method [33].

5. Conclusion and future directions

In this paper, we proposed a deep learning based approach for detection and localization of road accidents. The benefit of the proposed approach is that we need to only train the model on normal traffic events and anomalies are detected as out of distribution samples due to one-class learning paradigm. We assessed the performance of the model over three benchmarked datasets. As part of future direction, the code can be deployed on hardware like PYNQ for faster processing of video frames for anomaly detection. Moreover, research can be focused on the use of active learning for anomaly detection. Active learning would enable to update the model on the go according to newer behavior patterns found in both normal and anomalous data by having the human intervention.

CRediT authorship contribution statement

Karishma Pawar: Conception and design of study, Acquisition of data, Analysis and/or interpretation of data, Writing - review & editing. **Vahida Attar:** Conception and design of study, Analysis and/or interpretation of data, Writing - review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] A. Zanella, N. Bui, A. Castellani, L. Vangelista, M. Zorzi, Internet of things for smart cities, IEEE IoT J. 1 (1) (2014) 22–32.

[2] M. Kumar, K.S. Raju, D. Kumar, N. Goyal, S. Verma, A. Singh, An efficient framework using visual recognition for IoT based smart city surveillance, Multimedia Tools Appl. (2021) 1–19.

[3] K.K. Santhosh, D.P. Dogra, P.P. Roy, Anomaly detection in road traffic using visual surveillance: A survey, ACM Comput. Surv. 53 (2020) 1–26.

[4] W. Hu, X. Xiao, Z. Fu, D. Xie, T. Tan, S. Maybank, A system for learning statistical motion patterns, IEEE Trans. Pattern Anal. Mach. Intell. 28 (2006) 1450–1464.

[5] H. Tan, J. Zhang, J. Feng, F. Li, Vehicle speed measurement for accident scene investigation, in: 2010 IEEE 7th Int. Conf. E-Bus. Eng., 2010, pp. 389–392.

[6] F. Jiansheng, et al., Vision-based real-time traffic accident detection, in: Proceeding 11th World Congr. Intell. Control Autom., 2014, pp. 1035–1038.

[7] W. Sultani, J.Y. Choi, Abnormal traffic detection using intelligent driver model, in: 2010 20th Int. Conf. Pattern Recognit., 2010, pp. 324–327.

[8] A.R. Pathak, M. Pandey, S. Rautaray, Application of deep learning for object detection, Procedia Comput. Sci. 132 (2018) 1706–1717, http://dx.doi.org/10.1016/j.procs.2018.05.144.

[9] A.R. Pathak, M. Pandey, S. Rautaray, K. Pawar, Assessment of object detection using deep convolutional neural networks, vol. 673, 2018.

[10] G. Algan, I. Ulusoy, Image classification with deep learning in the presence of noisy labels: A survey, Knowl.-Based Syst. 215 (2021) 106771.

[11] K. Pawar, V. Attar, Deep learning approaches for video-based anomalous activity detection, World Wide Web 22 (2) (2019) 571–601, http://dx.doi.org/10.1007/s11280-018-0582-1.

[12] A.R. Pathak, M. Pandey, S. Rautaray, Topic-level sentiment analysis of social media data using deep learning, Appl. Soft Comput. 108 (2021) 107440.

[13] K. Pawar, V.Z. Attar, Assessment of autoencoder architectures for data representation, in: W. Pedrycz, S.-M. Chen (Eds.), Deep Learning: Concepts and Architectures, Springer International Publishing, 2020, pp. 101–132.

[14] A. Ouahabi, A. Taleb-Ahmed, Deep learning for real-time semantic segmentation: Application in ultrasound imaging, Pattern Recognit. Lett. 144 (2021) 27–34.

[15] S.M. Marvasti-Zadeh, L. Cheng, H. Ghanei-Yakhdan, S. Kasaei, Deep learning for visual tracking: A comprehensive survey, IEEE Trans. Intell. Transp. Syst. (2021).

[16] H. Ren, Y. Song, J. Wang, Y. Hu, J. Lei, A deep learning approach to the citywide traffic accident risk prediction, in: 2018 21st Int. Conf. Intell. Transp. Syst., 2018, pp. 3346–3351.

[17] M. Bortnikov, A. Khan, A.M. Khattak, M. Ahmad, Accident recognition via 3d cnns for automated traffic monitoring in smart cities, in: Sci. Inf. Conf., 2019, pp. 256–264.

[18] D. Tian, C. Zhang, X. Duan, X. Wang, An automatic car accident detection method based on cooperative vehicle infrastructure systems, IEEE Access 7 (2019) 127453–127463.

[19] Y. Yao, M. Xu, Y. Wang, D.J. Crandall, E.M. Atkins, Unsupervised traffic accident detection in first-person videos, 2019, ArXiv Prepr. arXiv:1903.00618.

[20] T. Ohgushi, K. Horiguchi, M. Yamanaka, Road obstacle detection method based on an autoencoder with semantic segmentation, in: Proc. Asian Conf. Comput. Vis., 2020.

[21] B. Varona, A. Monteserin, A. Teyseyre, A deep learning approach to automatic road surface monitoring and pothole detection, Pers. Ubiquitous Comput. 24 (2020) 519–534.

[22] S.S. Khan, M.G. Madden, A survey of recent trends in one class classification, in: L. Coyle, J. Freyne (Eds.), Artificial Intelligence and Cognitive Science, Springer Berlin Heidelberg, 2010, pp. 188–197.

[23] IITH Road Accident dataset, https://sites.google.com/site/dineshsinghindian/iith_accident-dataset.

[24] D. Singh, C.K. Mohan, Deep spatio-temporal representation for detection of road accidents using stacked autoencoder, IEEE Trans. Intell. Transp. Syst. 20 (3) (2019) 879–887, http://dx.doi.org/10.1109/TITS.2018.2835308.

[25] M. Naphade, S. Wang, D.C. Anastasiu, Z. Tang, M.-C. Chang, X. Yang, Y. Yao, L. Zheng, P. Chakraborty, C.E. Lopez, A. Sharma, Q. Feng, V. Ablavsky, S. Sclaroff, The 5th AI city challenge, in: IEEE Conf. Comput. Vis. Pattern Recognit. Work., 2021.

[26] Iowa DOT anomaly dataset, https://www.aicitychallenge.org/2021-data-and-evaluation/.

[27] Y. Zhao, W. Wu, Y. He, Y. Li, X. Tan, S. Chen, Good practices and a strong baseline for traffic anomaly detection, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2021, pp. 3993–4001.

[28] J. Wu, X. Wang, X. Xiao, Y. Wang, Box-level tube tracking and refinement for vehicles anomaly detection, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2021, pp. 4112–4118.

[29] J. Chen, G. Ding, Y. Yang, W. Han, K. Xu, T. Gao, Z. Zhang, W. Ouysang, H. Cai, Z. Chen, Dual-modality vehicle anomaly detection via bilateral trajectory tracing, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2021, pp. 4016–4025.

[30] K. Doshi, Y. Yilmaz, An efficient approach for anomaly detection in traffic videos, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2021, pp. 4236–4244.

[31] A. Aboah, A vision-based system for traffic anomaly detection using deep learning and decision trees, in: Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit., 2021, pp. 4207–4212.

[32] DoTA dataset, https://github.com/MoonBlvd/Detection-of-Traffic-Anomaly.

[33] Y. Yao, X. Wang, M. Xu, Z. Pu, E. Atkins, D. Crandall, When, where, and what? a new dataset for anomaly detection in driving videos, 2020, ArXiv Prepr. arXiv:2004.03044.

[34] Y.S. Chong, Y.H. Tay, Abnormal event detection in videos using spatiotemporal autoencoder, 2017, CoRR. abs/1701.0. http://arxiv.org/abs/1701.01546.

[35] M. Hasan, J. Choi, J. Neumann, A.K. Roy-Chowdhury, L.S. Davis, Learning temporal regularity in video sequences, in: Comput. Vis. Pattern Recognit. (CVPR), 2016 IEEE Conf., 2016, pp. 733–742.

[36] W. Liu, W. Luo, D. Lian, S. Gao, Future frame prediction for anomaly detection–a new baseline, in: Proc. IEEE Conf. Comput. Vis. Pattern Recognit., 2018, pp. 6536–6545.